AAR DRAFT · OPERATOR REVIEW

After-action review / NZ1 radio desk.

Exercise Noble Skywave 2026 · Station NZ1, Trentham, New Zealand · 20–22 October 2026.

15.5 points

Historical Back to the Future scoreboard: NZ1 ranked 90 at that saved snapshot. The final exercise total is not established here. [S2–S3]

304 RX jobs

Stored receive coordination jobs, including preparation records. These are workflow records, not 304 proven RF exchanges.

29 draft records

Reporting records cover 22 distinct callsigns. Submission flags, two-way flags and returned Web App outcomes need separate reconciliation.

PURPOSE & OBSERVED OUTCOME

Connect coordination to real radio evidence.

The desk brought main-chat requests, roster matching, frequency agreement, guarded voice calls, receive recordings and contact reporting into one console. Local evidence and saved scoring records show useful exchanges, alongside missed or unresolved opportunities.

Operational friction centred on actual receiver tuning, FLEX A/B alignment, antenna/tuner state, competing controllers, stale conversations and reporting follow-through. Queue invitations and completed TX audio repeatedly needed clearer separation from confirmed contacts. [S2, S4–S8]

Retain the improvements

Exact dial readback; one serialized RF pipeline; fresh listener acknowledgement; original audio; separate directional reports; visible reporting deadlines; manual and EMCON stops.

DECISIONS FOR THE NEXT ITERATION

Close the evidence loop faster.

Keep outreach broad while protecting each live RF exchange. Base “listening” and “transmitting” labels on attributable telemetry. Reconcile pending submissions with the other station and the official Web App.

Review status and scope

This is a new AAR draft compiled from saved local records. It has not been approved or submitted. Final official points, complete receipt coverage and a controlled antenna comparison remain outstanding.

Instructions schedule the competition from 20 October 1400Z to 22 October 2000Z. AAR points are due to the OIC by 20 November 2026. [S1, pp. 2, 5–6] All event times below are UTC unless labelled otherwise.

Snapshot captured: 2026-10-22T20:36:29.053700+00:00. The RX table includes preparation from 18 October; stored reporting data was last updated 2026-10-22T10:01:04.879921+00:00.

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Keep the two directions and outcomes distinct.

A transmission, a received recording, a peer report and an official score are different pieces of evidence.

RECORD / EVENT	WHAT THE RECORD SUPPORTS	RESULT / LIMIT
CA15 / US184 / PE2	Saved Web App ticker lists 6.0 / 5.0 / 4.5 points respectively: 15.5 total.	Historical scoring evidence, not a final total. [S2–S3]
US59 · 21 Oct	Saved operator-heard RX report on 10.9311 MHz USB with recording references.	RX evidence; the report file does not establish an official submission.
AU17 · 20 Oct 21:37Z	15.7331 MHz attempt aborted locally by the ANT2/SWR fault. Earlier receive window contained nothing readable.	NZ1→AU17 unresolved. “No joy” after the aborted TX is not a valid failed propagation result. [S4]
NCS · 20 Oct	Two calls completed on exactly 10.1950 MHz USB, with no confirmed NCS acknowledgement.	Separate two-call limit exhausted; unresolved NZ1 contact. NZ3’s successful 24.7000 MHz contact is external evidence only. [S5]
DE10 · 22 Oct 07:53Z	Completed 13.9531 MHz USB call: 20.8 s source audio, 42.6 W peak; unkey confirmed.	No new peer report established in the reviewed update. Later quiet RX does not prove DE10 transmitted. [S6]
PE9 / PE40 / JP1	Four durable reports marked TRANSMITTED_PEER_HEARD across these stations.	Peer-heard direction is recorded; these files explicitly have no confirmed readable NZ1 return.

Durable contact-report files

Seven files were found: four TX/peer-heard reports, two operator-heard RX reports, and one operator-reported two-way AU4 record. Their official_score_submitted flags are all false. These files are not the entire scoring ledger.

Historical attempts stay separate

AU17 also has an earlier 7.9001 MHz receive record; it does not resolve the later aborted 15.7331 MHz attempt. Later reports, altered budgets and new operating windows must not overwrite the meaning of an earlier event.

Call journal reviewed: 50 attempted-call entries, 31 marked transmitted, 12 result entries and one RX update. A transmitted flag can include an interrupted attempt; it is not a successful-contact count.

Operational lessons from the saved record.

Observed problems and implemented responses are retained here as lessons; the draft does not claim a fresh full-system retest.

AREA	OBSERVATION	RETAIN / IMPROVE
Exact dial / actual radio	Peers were told NZ1 was listening while the operator observed another frequency or mismatched A/B receivers.	Verify the assigned receiver’s native dial, mode and antenna before readiness; align paired receivers when required. [S2, S7]
Antenna / tuner faults	ANT2/SWR interrupted AU17’s call; later tuner recovery had to be verified before a new attempt.	Record local equipment aborts as incomplete tests. Obtain fresh readiness and run the guarded recovery path. [S4–S5]
Queue labels / real RF	CA48 appeared to be “Calling” after only a listening invitation.	Use Awaiting listening / Preparing TX / Transmitting / Receiving / Awaiting report, backed by actual attributable state. [S6, S10]
Competing control / restart	Overlapping desk and operator activity interrupted a DE10 attempt; startup EMCON and stale claims also blocked opportunities.	One controller and one RF lock; inspect fresh state after restart; respect manual ownership and explicit TX permission. [S2, S6–S8]
Outreach pace / stale work	Repeated proposals and old DMs consumed cycles while active stations were available in main chat.	Prioritize fresh requests; rotate quiet or completed peers; deduplicate invitations and expire stale offers. [S2, S6]
Recognition / report quality	Quiet or uncertain recordings, numeric reports and human hearing notes required different treatment.	Keep the original WAV and context; review uncertain text; never infer a two-way result from ASR confidence or a readiness reply. [S2, S8]

What worked

GRAFEX path advice helped replace poorly placed invitations with permitted candidate dials. Serialized TX offers, guarded return windows and the heartbeat/supervisor improved the workflow. The activity/map changes made the distinction between coordination and current RF clearer. [S2, S6, S8, S10]

A prediction assists selection; current operating permissions and explicit peer agreement still determine the usable dial. Public SDR or HackRF reception describes that receiver’s site, not what the other station heard.

Actions to close before the next deployment.

Proposed priorities and roles for operator assignment; these actions have not been marked complete by creating this report.

PRIORITY	ACTION	PROPOSED OWNER	ACCEPTANCE / NEXT STEP
P1	Reconcile official reporting	Operator / reporting lead	Resolve 12 pending responses and 12 records without structured results. Retain receipts and distinguish duplicates; verify the final scoreboard.
P1	Prove receiver/dial alignment	Radio / software leads	Demonstrate “listening” on the actual assigned receiver; compare native A/B readback, mode, antenna and capture context before announcement.
P1	Exercise ownership and recovery	Radio / software leads	Rehearse manual takeover, restart, stale claims and antenna faults. Verify only one RF owner, all stops effective, and unkey on every exit.
P1	Preserve event history	Software / evidence lead	Keep append-only TX/RX events across budget changes. Do not use the current session counter as a lifetime contact count.
P2	Make outreach timely	Desk operator / software lead	Work fresh main-chat and NCS-directed opportunities, rotate silent peers and ensure receive windows follow the actual agreement.
P2	Unify the reporting ledger	Software / reporting leads	Join RX/TX evidence, manual statements, draft flags and Web App outcomes. Prevent a submitted flag from presenting as confirmed scoring.
P2	Measure real performance	Test / radio leads	Run a shift-length exercise with time-stamped call-to-return timing, audio review, reconnect tests and matched antenna comparisons.
P2	Complete handover and AAR	Site lead	Review this draft, attach verified receipts and final results, assign owners, then send approved AAR points by the instructed deadline.

Recorded verification

The saved checklist deployment reports 256 backend checks, 118 focused checks, six frontend checks and a passing production build. Test groups may overlap and are not added into a single total. [S8]

Additional receive capability

The HackRF note records real USB IQ/audio on 9.9411 MHz USB, with audio capture and regression checks. This is receive-path evidence, not proof of an over-the-air station contact. [S9]

NCS’s separate exhausted two-call authorization and all frequency, power, antenna, SWR, EMCON, ownership and stop protections remain historical operating constraints. Preparing this AAR changes no live authorization.

Reporting records require reconciliation.

These are saved API outcomes and local flags, not a count of unique officially scored contacts.

3
success responses

2
exists responses

12
pending responses

12
no structured result

CALLSIGN	RECORDS	SAVED OUTCOMES
AU6	1	success
CA15	1	no structured result
CA46	1	pending
CA58	1	pending
CZ1	1	no structured result
IQ1	1	pending
JP1	3	exists / no structured result
PE2	2	no structured result
PE28	1	no structured result
PE40	3	no structured result / pending
PE47	1	pending

CALLSIGN	RECORDS	SAVED OUTCOMES
PE6	1	pending
PE9	1	pending
PT1	2	exists / pending
RO1	1	pending
US120	1	pending
US138	2	no structured result / success
US151	1	no structured result
US154	1	pending
US18	1	pending
US183	1	success
US184	1	no structured result

Flags are not receipts

All 29 local records say submitted=true; 20 say both_ways_confirmed=true and nine do not. Preserve the original rows and verify the underlying evidence. Do not count duplicate PT1, JP1, PE40 or US138 records as separate confirmed contacts.

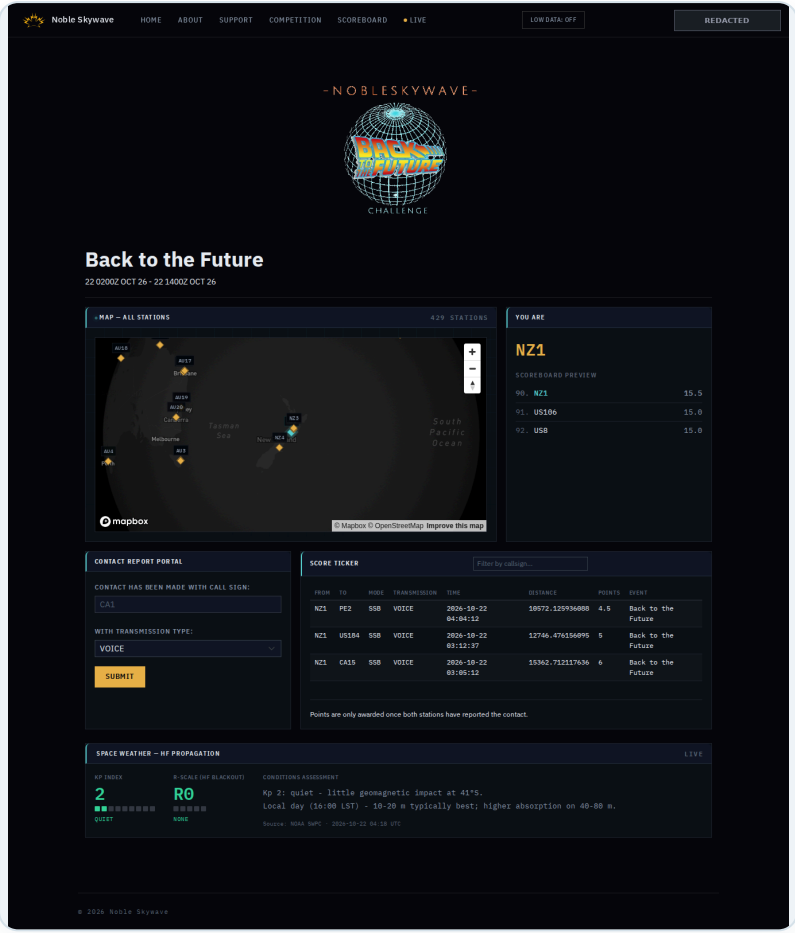
Receiver-job outcomes

304 stored jobs: 169 NOT_HEARD, 67 cancelled, 40 expired, 22 errors and 6 partial. These state counts describe workflow outcomes, not propagation failure rates.

“success” is saved for AU6, US183 and US138; “exists” for JP1 and PT1. No new official Web App lookup or submission was performed for this draft.

Source record and review boundaries.

The saved screenshot supports a historical scoring snapshot; it is kept separate from later local reporting records.



Original Desktop scoreboard image [S3], with the signed-in account name redacted. Its ticker shows PE2 4.5, US184 5.0 and CA15 6.0 points. It does not establish the final exercise result.

LOCAL SOURCES / EVIDENCE INDEX

[S1] Exercise instructions, pp. 2, 5–6
docs/20260624-3352-U-CFSCE_NOBLE_SKYWAVE_INSTRUCTIONS_2026.pdf

[S2] Saved session summary
Desktop/Session_Summary_20261022_161308Z.txt

[S3] Historical Web App scoreboard
Desktop/NZ1_BackToTheFuture_scoreboard_20261022.png

[S4] AU17 directional status
docs/AU17-STATUS-2026-10-21.md

[S5] NCS calls and monitor handover
docs/NCS-SETUP-2026-10-21.md

[S6] Queue and DE10 evidence
docs/QUEUE-STATUS-2026-10-22.md

[S7] Native radio-state audit
docs/RADIO-DESK-AUDIT-2026-10-22.md

[S8] Checklist deployment and validation
docs/DESK-CHECKLIST-DESKTOP-SUMMARY-2026-10-22.txt

[S9] HackRF receive verification
docs/HACKRF-RECEIVE-2026-10-22.md

[S10] Live map and status verification
docs/MAP-LIVE-ACTIVITY-2026-10-22.md

Retained with this AAR

A read-only snapshot of the receive-state totals, reporting records, durable report metadata, call-journal counts and source hashes is stored beside the PDF. The snapshot excludes simulation fixtures used in the separate workflow showcase.

Review still required: final official score and receipts; station-report reconciliation; complete contact coverage across stores; controlled antenna comparison; operator approval and submission. No final ranking, antenna winner or RF success percentage is inferred.